

# Michael Crawshaw

Ph.D. Student

George Mason University

Research Interests: Optimization for machine learning, Distributed optimization

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🌐 [Personal Website](#)

🔗 [Google Scholar](#)

## EDUCATION

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### George Mason University

Ph.D. in Computer Science

2019 - Present

M.S. in Computer Science

2019 - 2022

Advisor: Mingrui Liu

### The Ohio State University

2015 - 2019

B.S. in Mathematics and Computer Science, Honors

## PUBLICATIONS

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### Constant Stepsize Local GD for Logistic Regression: Acceleration by Instability

Michael Crawshaw, Blake Woodworth, Mingrui Liu

International Conference on Machine Learning, 2025.

### Local Steps Speed up Local GD for Heterogeneous Distributed Logistic Regression

Michael Crawshaw, Blake Woodworth, Mingrui Liu

International Conference on Learning Representations, 2025.

### Complexity Lower Bounds of Adaptive Gradient Algorithms for Non-convex Stochastic Optimization under Relaxed Smoothness

Michael Crawshaw, Mingrui Liu

International Conference on Learning Representations, 2025.

### Federated Learning under Periodic Client Participation and Heterogeneous Data: A New Communication-Efficient Algorithm and Analysis

Michael Crawshaw, Mingrui Liu

Conference on Neural Information Processing Systems, 2024.

### Provable Benefits of Local Steps in Heterogeneous Federated Learning for Neural Networks: A Feature Learning Perspective

Yajie Bao, Michael Crawshaw, Mingrui Liu

International Conference on Machine Learning, 2024.

### Federated Learning with Client Subsampling, Data Heterogeneity, and Unbounded Smoothness: A New Algorithm and Lower Bounds

Michael Crawshaw\*, Yajie Bao\*, Mingrui Liu (\* denotes equal contribution)

Conference on Neural Information Processing Systems, 2023.

### EPISODE: Episodic Gradient Clipping with Periodic Resampled Corrections for Federated Learning with Heterogeneous Data

Michael Crawshaw, Yajie Bao, Mingrui Liu

International Conference on Learning Representations, 2023.

### Robustness to Unbounded Smoothness of Generalized SignSGD

(Alphabetical order) Michael Crawshaw, Mingrui Liu, Francesco Orabona, Wei Zhang, Zhenxun Zhuang

Neural Information Processing Systems, 2022.

### Fast Composite Optimization and Statistical Recovery in Federated Learning

Yajie Bao, Michael Crawshaw, Mingrui Liu

International Conference on Machine Learning, 2022.

## EMPLOYMENT

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### Flatiron Institute

May 2025 - August 2025

Summer Predoctoral Researcher, Center for Computational Mathematics

- Developed novel non-Euclidean gradient descent methods to train deep neural networks, especially language models.
- Proposed methods match or outperform state-of-the-art with significantly improved robustness to hyperparameter tuning.

### Olive

March 2018 - August 2019

Machine Learning Engineering Intern

- Developed computer vision functionality for desktop automation software with applications to healthcare operations.
- Trained deep neural networks for object detection with various techniques, including Faster R-CNN and DARTS.

## AWARDS

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|--------------------------------------------------------------------------------------------|-------------|
| <b>Top Reviewer</b> , ICML                                                                 | 2025        |
| <b>Best Reviewer</b> , AISTATS                                                             | 2025        |
| <b>Top Reviewer</b> , NeurIPS (top 8%)                                                     | 2023        |
| <b>Institute for Digital Innovation Predoctoral Fellowship</b> , George Mason University   | 2022        |
| <b>NSF XSEDE startup allocation</b> , National Science Foundation                          | 2020        |
| <b>Summer Ph.D. Research Initiation Award</b> , George Mason University                    | 2020        |
| <b>Outstanding Graduate Teaching Assistant</b> , George Mason University                   | 2020        |
| <b>Gordon Memorial Fund Scholarship</b> , The Ohio State University                        | 2017 - 2019 |
| <b>Honorable Mention, Raser-Bareis-Gordon Math Competition</b> , The Ohio State University | 2017        |
| <b>7th Place, FIRST Tech Challenge World Competition</b> , FIRST                           | 2015        |

## ACADEMIC SERVICE

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| <b>Reviewer</b> , ICLR, AISTATS (awarded Best Reviewer), ICML (awarded Top Reviewer) | 2025 |
| <b>Reviewer</b> , AISTATS, ICML, NeurIPS                                             | 2024 |
| <b>Reviewer</b> , NeurIPS (awarded Top Reviewer, top 8%)                             | 2023 |

## TEACHING

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|---------------------------------------------------------------------|-------------------------------------|
| <b>Graduate Teaching Assistant</b> , George Mason University        | 2019-2022                           |
| CS 657: Mining Massive Datasets                                     | Fall 2020, Fall 2021                |
| CS 471: Operating Systems                                           | Fall 2020, Fall 2021, Spring 2022   |
| CS 583: Analysis of Algorithms                                      | Spring 2021                         |
| CS 571: Operating Systems                                           | Spring 2021                         |
| CS 330: Formal Methods and Models                                   | Fall 2019, Spring 2020              |
| <b>Undergraduate Teaching Assistant</b> , The Ohio State University | 2017 - 2018                         |
| CSE 3321: Automata and Formal Languages                             | Summer 2017, Fall 2017, Spring 2018 |
| <b>Undergraduate Honors Math Mentor</b> , The Ohio State University | 2016 - 2017                         |
| Math 4181H: Honors Analysis I                                       | Fall 2016                           |
| Math 4182H: Honors Analysis II                                      | Spring 2017                         |

## RELEVANT COURSEWORK AND SKILLS

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**GMU Coursework:** Optimization for machine learning, deep learning, computer vision, theory of computation, algorithms, graphics, software testing.

**OSU Math Coursework:** Real analysis, linear algebra, differential equations, probability, statistics, combinatorics, complex analysis, number theory, abstract algebra.

**OSU CS Coursework:** Software, digital logic, databases, operating systems, networking, theory of computation, machine learning, neural networks, natural language processing, GPU programming (CUDA).

**Programming:** Python (PyTorch, TensorFlow), Bash, Java, C, Git, Latex